Inside - Outside Game

Number the members of your group, #1, #2, #3, #4.

Round 1: Numbers 1 and 2 will randomly select a number between 1 and 20.
Number 1 will use his/her number to find the inside function on the List of Functions below.
Number 2 will use his/her number to get the outside function from the list.
Number 3 will write the composite function on the board.
Number 4 will find its derivative. Don't worry about simplifying your answer. Other members of the group will check #4's work.

Round 2: Rotate positions. #2 gets the inside function, #3 gets the outside function, #4 writes the composite function, #1 finds the derivative. Other members will check #1's work.

Round 3: Rotate again. #3 gets the inside function, #4 gets the outside function, #1 writes the composite function, #2 finds the derivative. Other members will check #2's work.

Round 4: Rotate again. #4 gets the inside function, #1 gets the outside function, #2 writes the composite function, #3 finds the derivative. Other members will check #3's work.

You may do as many rounds as you like, but you must do at least 4.

1. $y = 3x + 4$
2. $y = x^2 + 1$
3. $y = \sqrt{2x - 5}$
4. $y = \dfrac{1}{x}$
5. $y = (x - 5)^4$
6. $y = \sqrt[3]{x^2}$
7. $y = x^4 - 1$
8. $y = \dfrac{x}{x - 3}$
9. $y = 2x^3 - 10x^2 + 2x$
10. $y = \sqrt{\dfrac{x - 4}{2}}$
11. $y = \sqrt{8 - x}$
12. $y = (2x - 1)(x + 5)$
13. $y = 5x^3$
14. $y = 1 - \dfrac{3}{x^4}$
15. $y = 2 - \sqrt{x}$
16. $y = 27 - x^3$
17. $y = x^2 - x - 12$
18. $y = \sqrt[3]{4 - x^2}$
19. $y = 5x + 7$
20. $y = \dfrac{x + 1}{x - 2}$